Week 4 | **CLO** 3-B · 3-C · 3-D (R) | **Before** In-Class Quiz I | **File** 5 / 8

3-4 Exploratory Data Analysis

التحليل الاستكشافي للبيانات — الملخص الخماسي والمخطط الصندوقي وقراءة الشكل والمقارنة

1 · Traditional statistics vs EDA

AMANI · HADHRAMI TABLE 3-5

ASPECT	TRADITIONAL STATISTICS	EXPLORATORY DATA ANALYSIS (EDA)
Data organisation	frequency distribution	**stem-and-leaf plot**
Main graphs	histogram, ogive	**boxplot** (box-and-whisker plot)
Measure of centre	mean	**median**
Measure of variation	standard deviation	**IQR = $Q_3 - Q_1$**
Purpose	test specific hypotheses or confirm assumptions from collected data, e.g. has the average income increased compared with last year?	explore the data and find patterns, trends or outliers **without prior assumptions**

الشرح بالعربي — لماذا التحليل الاستكشافي؟

الإحصاء التقليدي يبدأ بفرضية ويختبرها باستخدام المتوسط والانحراف المعياري. أما التحليل الاستكشافي فيبدأ **بلا افتراضات**: ننظر إلى البيانات لنكتشف شكلها وقيمها الشاذة. ولذلك يستخدم **الوسيط والمدى الربيعي** بدل المتوسط والانحراف، لأنهما **لا يتأثران بالقيم الشاذة** التي نبحث عنها أصلًا. احفظ الجدول بأعمدته الثلاثة، فهو سؤال تعريف مباشر في الكويز.

2 · Five-number summary and boxplot

BOTH

A data set is summarised by five values, reported **in this order**: (1) minimum; (2) $Q_1 = P_{25}$; (3) $Q_2 = \text{median} = P_{50}$; (4) $Q_3 = P_{75}$; (5) maximum.

Constructing a boxplot AMANI: **Step 1** find the five-number summary. **Step 2** draw a horizontal axis with a scale that includes the minimum and maximum. **Step 3** draw a box whose vertical sides pass through Q_1 and Q_3, with a vertical line through the median. **Step 4** draw a line (whisker) from the minimum to the left side of the box and from the maximum to the right side.

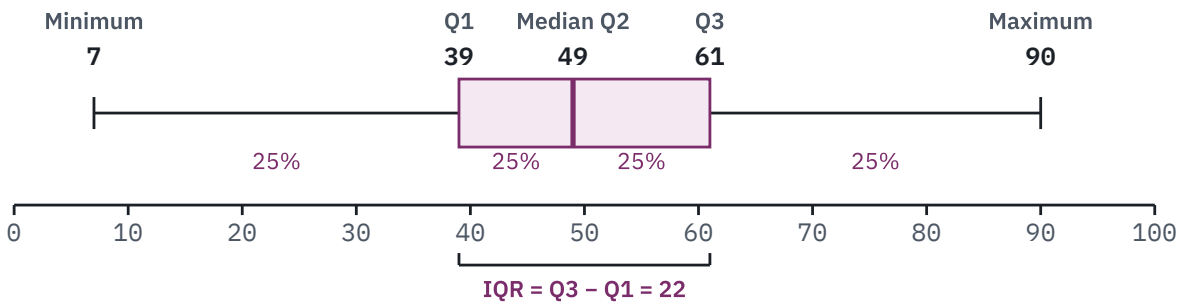
SIMPLE EXAMPLE AMANI

7, 15, 36, 39, 42, 43, 46, 49, 50, 51, 52, 61, 69, 80, 90. Construct a boxplot.

n = 15 (odd), so the median is the 8th value, **49**. Lower seven: 7 15 36 **39** 42 43 46, so Q_1 = **39**. Upper seven: 50 51 52 **61** 69 80 90, so Q_3 = **61**.

Five-number summary: **7, 39, 49, 61, 90**.

The appearance of a boxplot changes with skewness, but its structure stays the same.



Five-number summary 7, 39, 49, 61, 90. Each of the four pieces (left whisker, two halves of the box, right whisker) holds 25% of the data, however long it looks.

الملخص الخماسي 7، 39، 49، 61، 90. كل جزء من الأجزاء الأربعة (الخط الأيسر، نصفا الصندوق، الخط الأيمن) يحتوي 25% من البيانات مهما كان طوله.

ACTIVITY AMANI

Construct a boxplot for 12, 5, 22, 30, 7, 36, 14, 42, 15, 53, 25.

Ordered: 5 7 12 14 15 **22** 25 30 36 42 53 (n = 11). MD = **22**. Q_1 = median of 5 7 12 14 15 = **12**. Q_3 = median of 25 30 36 42 53 = **36**.

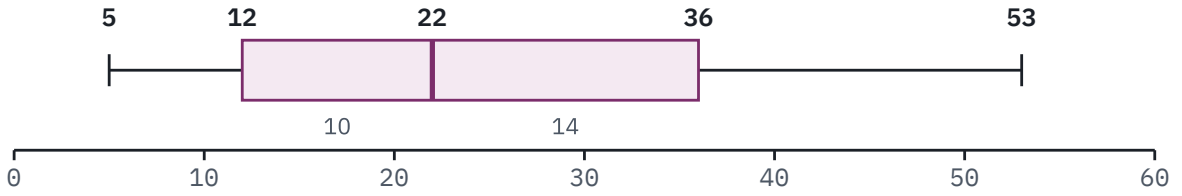
Five-number summary: **5, 12, 22, 36, 53**. IQR = 24 and the fences are −24 and 72, so there are no outliers.

EX 3-37 · METEORITES IN 10 U.S. STATES HADHRAMI

89, 47, 164, 296, 30, 215, 138, 78, 48, 39

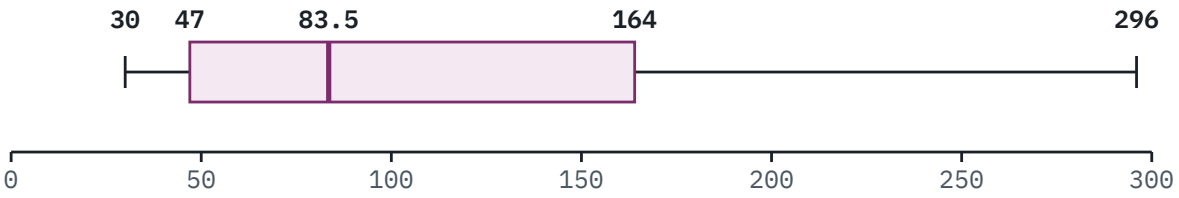
Ordered: 30 39 47 48 78 89 138 164 215 296. MD = (78 + 89)/2 = **83.5**. Q_1 = **47**, Q_3 = **164**; min 30, max 296.

Five-number summary: **30, 47, 83.5, 164, 296**.



Activity solution. The median sits left of the box centre (10 vs 14 on each side) and the right whisker (17) is longer than the left (7): the data are positively (right) skewed.

حل النشاط: الوسيط يقع يسار منتصف الصندوق (10 مقابل 14)، والخط الأيمن (17) أطول من الأيسر (7)، إذن التوزيع ملتوٍ نحو اليمين (موجب).



Example 3-37. The median is close to Q1 and the right whisker is far longer than the left: positively (right) skewed.

مثال 3-37: الوسيط قريب من Q1 والخط الأيمن أطول بكثير من الأيسر، إذن التوزيع ملتوٍ نحو اليمين.

3 · Reading shape from a boxplot BOTH · CLO3-B

SHAPE	MEDIAN INSIDE THE BOX	WHISKERS (LINES)
Symmetric	near the centre	about the same length
Positively (right) skewed	left of centre	right line longer
Negatively (left) skewed	right of centre	left line longer

HOW TO ANSWER "DESCRIBE THE SHAPE" FOR FULL MARKS

Quote the evidence with numbers: "The median (83.5) is closer to Q_1 (47) than to Q_3 (164): 36.5 versus 80.5. The right whisker (132) is much longer than the left (17). So the distribution is **positively skewed**." Dr. Amani's notes join the two clues with AND. If they disagree, say so and give more weight to the median position.



Left (negatively) skewed
median right · left whisker longer

Symmetric
median centred · whiskers equal

Right (positively) skewed
median left · right whisker longer

Read skewness from two clues: where the median sits inside the box, and which whisker is longer. Both point toward the long tail.

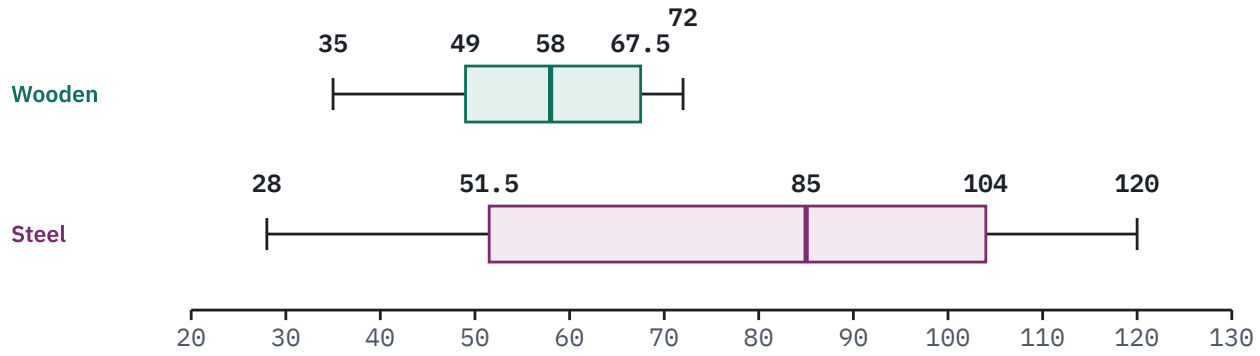
نقرأ الالتواء من دليلين: موقع الوسيط داخل الصندوق، وأي الخطين أطول. وكلاهما يشير إلى جهة الذيل الطويل.

4 · Side-by-side boxplots BOTH · CLO3-C

EX 3-38 · ROLLER COASTER SPEEDS (MPH) HADHRAMI

Wood: 50 56 60 48 35 67 72 68 · Steel: 55 70 48 28 100 106 102 120. Compare the distributions using boxplots.

Wood ordered: 35 48 50 56 | 60 67 68 72, giving five-number summary **35, 49, 58, 67.5, 72**. Steel ordered: 28 48 55 70 | 100 102 106 120, giving **28, 51.5, 85, 104, 120**.



Example 3-38 on one scale. Steel coasters have a much higher median (85 vs 58), a much larger IQR (52.5 vs 18.5) and range, and are negatively skewed; wooden coasters are about symmetric.

مثال 3-38 على مقياس واحد: الوسيط للحديدية أعلى بكثير 85) مقابل (58، ومداها الربيعي ومداها أكبر بكثير، وتوزيعها ملتوٍ لليسار، بينما الخشبية متماثلة تقريبًا.

(1) The median of the steel coasters is much higher (85 vs 58). (2) The IQR (spread) of the steel coasters is much larger (52.5 vs 18.5). (3) The range of the steel coasters is larger (92 vs 37). (4) The steel coasters are **negatively skewed** (median nearer Q_3); the wooden coasters are about symmetric.

COMPARING TWO CLASSES' GRADES AMANI

The same four-point comparison: Class B's **median** is higher (it performed better overall); Class B's **IQR** is larger (grades more spread out); its **range** is wider (more variation in performance); Class B is **left-skewed** (some students scored much lower than the median) while Class A is **right-skewed**. Use this template, *centre → spread (IQR) → range → shape*, for every side-by-side question.

الشرح بالعربي — قراءة المخطط الصندوقي والمقارنة

المخطط الصندوقي يُرسم من **الملخص الخماسي**: أصغر قيمة، Q_1 ، الوسيط، Q_3 ، أكبر قيمة. وكل جزء من أجزائه الأربعة يحتوي **25%** من البيانات مهما كان طوله، فالجزء الأطول لا يعني بيانات أكثر بل **تشتتًا أكبر**. لمعرفة الشكل: إذا كان الوسيط **يسار** منتصف الصندوق والخط **الأيمن** أطول ← ملتوٍ لليمين (موجب)، والعكس ← ملتوٍ لليسار (سالب). وعند المقارنة بين مجموعتين اكتب أربع نقاط بالترتيب: (1) الوسيط، (2) المدى الربيعي، (3) المدى، (4) الشكل. ودعّم كل نقطة **بالأرقام**.

R FOR LAB QUIZ I (OUTLINE CLO3-D)

`fivenum(x)` gives the five-number summary, `boxplot(x, horizontal = TRUE)` draws the boxplot, and `boxplot(wood, steel, names = c("Wood", "Steel"), horizontal = TRUE)` draws side-by-side boxplots. R's boxplot whiskers stop at the last value inside the 1.5·IQR fences and draw any outliers as separate points. `fivenum()` can differ from the hand method when n is odd: for Dr. Amani's 15 values it gives hinges 40.5 and 56.5 instead of $Q_1 = 39$ and $Q_3 = 61$.

Chapter 3 study set: 8 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's Spring 2024 slides, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

00 · Exam brief: what the outline tests · 01 · Lesson 3-1: Measures of Central Tendency · 02 · Lesson 3-2: Measures of Variation · 03 · Lesson 3-3: Measures of Position · **04 · Lesson 3-4: Exploratory Data Analysis** · 05 · Revision pack: formulas, decisions, traps, R · 06 · Exam-style questions (no answers) · 07 · Full worked answers